VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama", Belagavi, Karnataka, INDIA

An Assignment-Based Project Report
for the course

"MEDICAL INSURANCE COST PREDICTION SYSTEM"

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering in
Computer Science and Engineering

Submitted By

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2025– 2026

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CERTIFICATE

Certified that the report entitled "MEDICAL INSURANCE COST PREDICTION SYSTEM" is a bonafide record of the assignment-based project work carried out by PRAJWAL MURGOD, bearing USN 2JH23CS097, as a part of the academic requirements for the course Machine Learning (BCS602) in the VI Semester of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi, during the academic year 2025–2026. This work includes the study and implementation of various concepts and applications related to Blockchain Technology. It is certified that this report has been prepared under the guidance of the faculty and has been submitted in partial fulfillment of the course requirements.

Prof. Harshita C K Assistant Professor Dept. of CSE JCET, Hubballi	Dr. Maheshkumar Patil, HOD Dept. of CSE JCET, Hubballi

Name of the Examiners	Signature with date
1. _____	_____

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DECLARATION

I, PRAJWAL MURGOD, bearing USN 2JH23CS097,student of sixth Semester B.E, Department of Computer Science and Engineering, Jain College of Engineering and Technology, Sai Nagar, Hubballi, declare that the Blockchain Project entitled "MEDICAL INSURANCE COST PREDICTION SYSTEM" has been carried out by us and submitted in partial fulfillment of the course requirements for the award of degree in Bachelor of Engineering in Computer Science and Engineering from Visvesvaraya Technological University, Belagavi during the academic year 2025-2026. The matter embodied in this report has not been submitted to any other university or institution for the award of any other degree.

PRAJWAL MURGOD 2JH23CS097

Place: Hubballi
Date: 29/04/26

ABSTRACT

The rapid escalation of healthcare costs has made it increasingly important to predict medical insurance expenses accurately. This project presents a Medical Insurance Cost Prediction System, which combines statistical analysis, machine learning, and modern web technologies to deliver real-time cost predictions based on user inputs.

The system utilizes a structured dataset containing demographic and behavioral attributes such as age, gender, body mass index (BMI), smoking habits, number of dependents, and geographic region. Extensive exploratory data analysis (EDA) was conducted using Pandas, Seaborn, and Matplotlib to identify patterns, detect outliers, and understand feature relationships.

A Multiple Linear Regression model was implemented using Scikit-learn. The model achieved an R^2 score of approximately 0.75, demonstrating strong predictive capability while maintaining interpretability.

The trained model was deployed via a RESTful API using Flask, and a user-friendly frontend was developed using Next.js with Tailwind CSS. The application supports region mapping, currency conversion, and an informational interface on insurance providers.

This project demonstrates how machine learning can be effectively integrated into a full-stack system to provide practical, scalable, and user-centric solutions in the healthcare domain..

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CHAPTER 1

INTRODUCTION

Background

The healthcare industry generates vast amounts of data, making it an ideal domain for applying machine learning techniques. Insurance providers rely on predictive models to estimate risk and determine premiums. However, these models are often opaque and inaccessible to end users.

This project bridges the gap by providing a transparent, interactive system that allows users to estimate their medical insurance costs based on personal attributes.

Problem Statement

Medical insurance costs vary widely due to multiple interacting factors. Traditional estimation methods are either manual, inaccurate, or inaccessible. There is a need for a system that:

- Accurately predicts insurance costs
- Provides real-time user interaction
- Is interpretable and scalable

Objectives

- Perform statistical analysis to understand insurance cost drivers
- Develop a robust regression-based predictive model
- Deploy the model through a scalable backend API
- Build an intuitive and responsive frontend interface
- Enhance usability through additional features like currency conversion

Scope of the Project

The system focuses on:

- Predictive modeling using structured tabular data
- Web-based interaction for real-time predictions
- Deployment-ready architecture

Limitations include:

- Use of a relatively small dataset
- Linear model assumptions
- No deep learning or real-time insurance data integration

CHAPTER 2
IMPLEMENTATION

1. Data Analysis and Preprocessing

Dataset Overview

The dataset consists of 7 attributes:

Feature	Type	Description
Age	Numerical	Age of the individual
Sex	Categorical	Gender
BMI	Numerical	Body Mass Index
Children	Numerical	Number of dependents
Smoker	Categorical	Smoking status
Region	Categorical	Residential region
Charges	Numerical	Insurance cost

Exploratory Data Analysis (EDA)

EDA revealed several important insights:

- Right-skewed distribution of insurance charges
- Strong correlation between smoking and high charges
- Moderate correlation between age and charges
- Presence of outliers in BMI and charges

Visualization tools used:

- Histograms → distribution analysis
- Boxplots → outlier detection
- Scatterplots → feature relationships

Feature Engineering

- Categorical encoding using label encoding
- No missing values detected
- Feature scaling not required for linear regression

2. Machine Learning Model

Mathematical Model

The Multiple Linear Regression model is defined as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$$

Where:

- y= predicted insurance cost
- x_i= independent variables
- β_i= coefficients
- ϵ= error term

Training Process

- Dataset split: 80% training, 20% testing
- Model trained using Ordinary Least Squares (OLS)
- Implemented using Scikit-learn

Model Evaluation Metrics

- **R^2 Score:** ~0.75
- **Mean Absolute Error (MAE):** Measures average prediction error
- **Mean Squared Error (MSE):** Penalizes larger errors

Interpretability

The model provides insight into feature importance:

- Smoking status → highest impact
- BMI → moderate impact
- Age → gradual increase effect

3. Backend API (Flask)

The backend system was implemented using Flask.

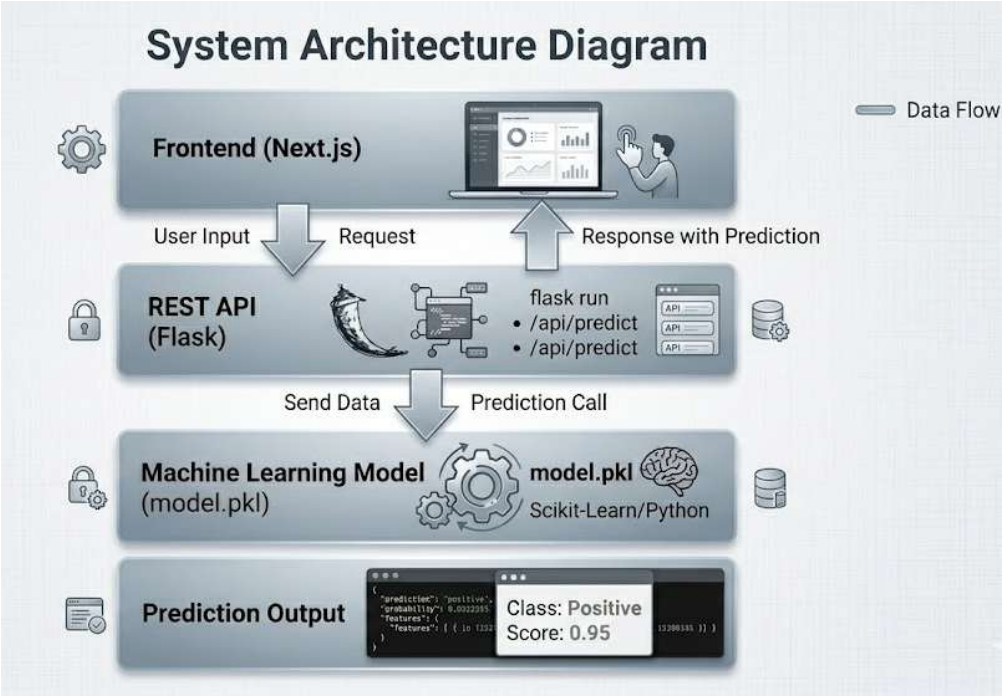


Fig 2.1: System Architecture

Key Functionalities

- Input validation
- Data preprocessing
- Model inference
- JSON response handling
- Cross-Origin Resource Sharing (CORS)

Error Handling

- Invalid inputs handled gracefully
- HTTP status codes used for communication
- Logging for debugging

4. Frontend Application

The frontend was built using Next.js and styled with Tailwind CSS.

User Interface Design

- Glassmorphism UI design
- Dark mode support
- Responsive layout

Core Features

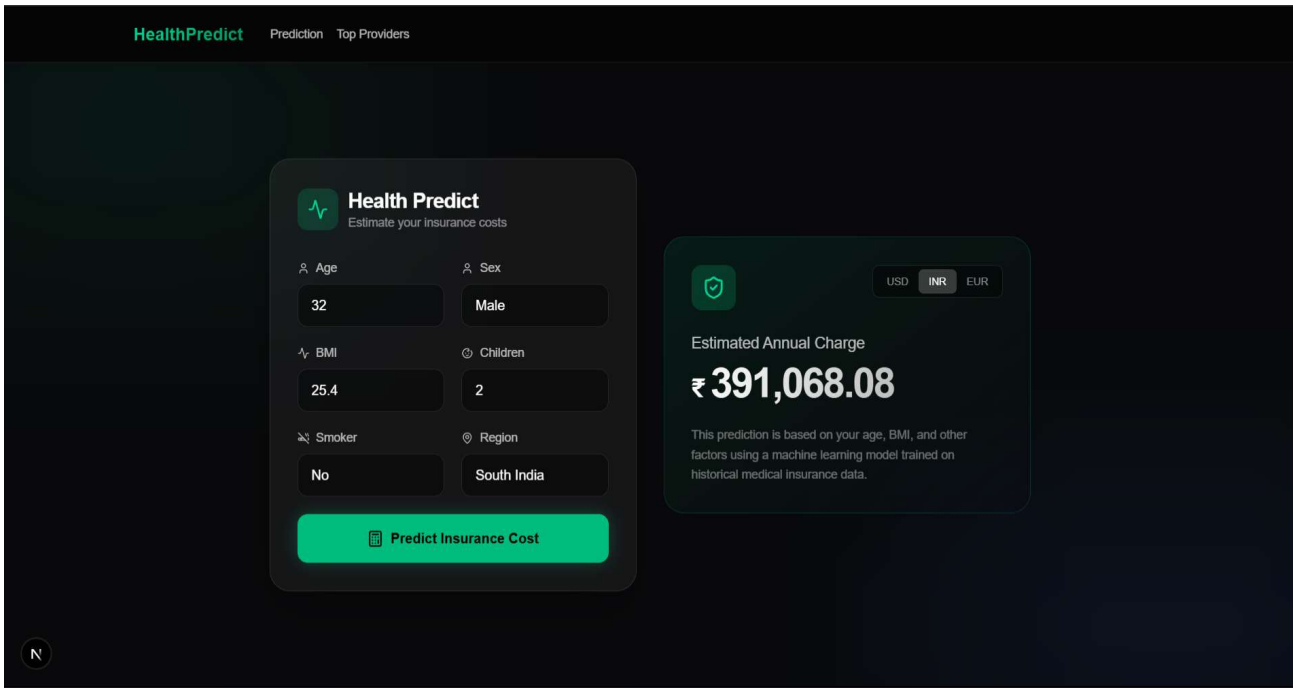
1. **Prediction Form**
   User inputs health and demographic data
2. **Real-Time Prediction**
   Displays predicted insurance cost instantly
3. **Currency Converter**
   Converts output into INR, USD, EUR
4. **Region Mapping**
   Maps Indian regions to dataset categories
5. **Top Providers Page**

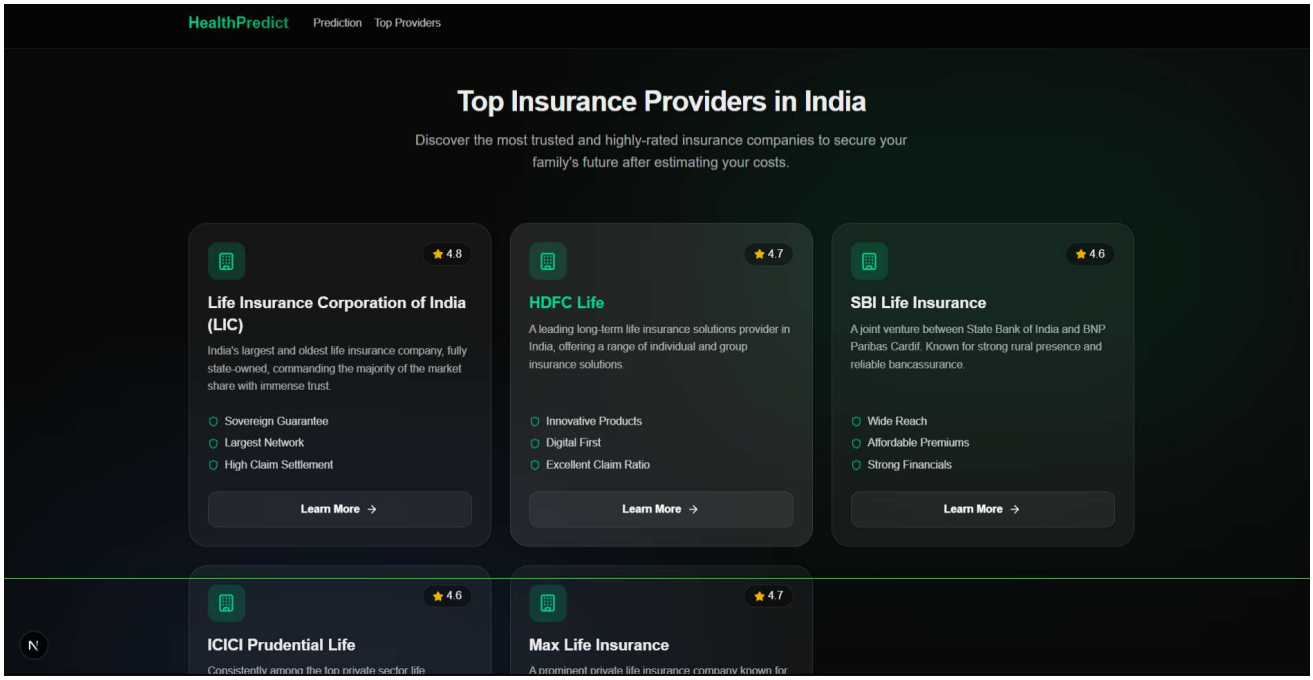
Frontend-Backend Communication

- API calls using fetch/axios
- JSON data exchange
- Asynchronous request handling

CHAPTER 3

RESULTS





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